Biospectra : Vol.8(1), March, 2013, pp 119-124
An International Biannual Refereed Journal of Life Sciences

Environmental Science

ISSN : 0973-7057

Catalytic and enzymatic activities of copper in human body-an overview

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Received , 15th January, 2013 ;Revised: 15th February, 2013

Abstract :Biochemical and crystallographic evidence of copper and its complexes are necessary for structural and catalytic properties of cuproenzymes**.**The resisting power of the body is increased by oral administration of Schiff base copper complex. The fundamental role of copper and the recognition of its complex are used as important bioactive compound and potential drug for therapeutic intervention in various diseases.Copper deficiency is usually the consequenceof decreased copper stores at birth, inadequate dietary copper intake, poor absorption, and elevated requirements induced by rapid growth, or increased copper losses. The most frequent clinical manifestations of copper deficiency are anemia, neutropenia, and bone abnormalities. Recommendations for dietary copper intake and total copper exposure, including the potable water, should consider that copper is an essential nutrient with potential toxicity if the load exceeds tolerance.

Key words : Catalytic, enzymatic, copper and bioactive.

INTRODUCTION

Schiff base ligand usually contains both N and O donor atoms although purely N as well as N,S donor exists. The chemistry of copper complexes with schiff base ligand and their application have aroused considerable interest because of preparative accessibility, diverse reactivity and structural variability. To extend the knowledge with respect to the coordination and biological properties of the copper metal complex of the Schiff base, we describe here the synthesis and characterization of Cu (II) complex Schiff base. Artificial receptors of biologically active molecules have attracted attention from the view point of molecular recognition .Because derivatives of barbituric acid are important member of pyridine family , they are suitable for this purpose ,but their biological activity is mainly attributed to tautomerisation,acid-base equilibrium, and to the nature of their substituent. For example, many scientists in recent years have shown that the related merocyanin dyes are effective sensitizers with interesting photo physical properties. Schiff base are generally bidentate, tridentate tetra dentate or polydentate ligand capable of forming very stable complexes with transition metal .Schiff base derived from aromatic amines and aromatic aldehydes have a wide variety of application in many fields e.g.biological ,inorganic and analytical chemistry. Application of many new analytical devices requires the presence of organic reagents as essential compound of the measuring system. Schiff bases are used e.g. in optical and electrochemical sensor, as well as in various chromatographic methods, to enable detection of enhanced selectivity and sensitivity. Among the organic reagents actually used, Schiff base possess excellent characteristics, structural similarity with natural biological substances, relatively simple preparation procedures and the synthetic flexibility they enable design of suitable structural properties. Copper is a transition metal with three oxidation states: Cu^0,$Cu^{1+,}$ and $Cu^{2+.}$ The cupric state is found most often in biological systems. A need for copper in higher animals has been suggested because hemocyanin, the oxygen-carrying molecule of invertebrates, was found to contain copper, but the potentiales sentiality of copper for humans was not recognized until 1928when Hart et al[1] showed copper to

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be essential for erythropoiesis in rats fed a milk-based diet. They were able to correct the anemia by adding ash from animal or vegetable sources to the diet. They went on to show that the hydrogen sulfide precipitate from the ash, which contained copper sulfide, was responsible for the recovery. Similar findings in humans established the basis for essentiality[1]. This paper will review present knowledge on the essential nature of copper for humans, the biochemical basis for essentiality, the adequacy of the dietary copper supply, factors that condition deficiency, and the special conditions that affect copper nurture in early infancy. Organisms as diverse as yeast and mammals share a requirementfor the regulation of copper metabolism to ensure correct function of several copper-binding proteins. Recently, biochemical and molecular studies have provided new evidence for homeostatic cellular processes for copper, contributing to a better understanding of the molecular mechanisms involved in this regulation. Copper affects enzyme activity, both as a cofactor and as an allosteric component of several cuproenzymes. In addition, important aspects of the copper-dependent regulatory mechanisms in genetic expression of different target genes have been found using yeast as a eukaryotic model.

MATERIALAND METHODS

Preparation of unsymmetrical Schiff bases

Ethanolic solutions of 2-hydroxy-1-naphthaldehyde (3.444 g, 20 mmol), 1,2-phenylenediamine (2.163 g, 20 mmol) and salicylaldehyde or substituted salicylaldehyde (20 mmol) in absolute ethanol (75 mL each) were prepared and chilled in the refrigerator at 4^0C for 15 min. To a stirred solution of the cold 2-hydroxy-1-naphthaldehyde, cold solution of 1, 2-phenylenediamine was added dropwise followed by the addition of cold solution of salicylaldehyde (or substituted salicylaldehyde) over a period of 2 min. The mixture was kept stirred at room temperature for 4 days, after which the mixture was warmed to, and kept at, 70^0Cfor 20 min with stirring to dissolve any unreacted reactants and to complete the reaction. The product was filtered hot and washed twice with ice cold ethanol. The orange product obtained was purified by digesting in hot ethanol, filtered hot, and dried in a dessicator over silica gel.

RESULT AND DISCUSSION

Studies on the biochemical basis of copper essentiality have shown that an important number of copper-containing proteins display oxidative reductase activity . Copper functions as an electron transfer intermediate in redox reactions. Copper has been shown to be an essential cofactor for catalytic activity in protein lysine 6-oxidase and catechol oxidase. In addition, crystallographic studies of Cu/Zn superoxide dismutase 1(SOD1), cytochrome-*c* oxidase, and ceruloplasmin confirmed that copper is an essential cofactor in oxidative and reductase enzymes .Cytochrome-*c* oxidase is an inner-mitochondrial–membrane protein complex that catalyzes the reduction of molecular oxygen to water and utilizes the free energy of this reaction to generate a transmembrane proton gradient during respiration.Ceruloplasmin is a blue multicopper oxidase that contains > 65% of the copper found in serum in vertebrate species. The crystal structure of ceruloplasmin confirms the presence of six tightly bound copper ions, three of them forming type I copper centers involved in electron transfer processes. The other three copper ions are in a single trinuclear center, which is the oxygen-activating site during the catalytic cycle of the enzyme. SOD1 is localized in the cytoplasm, where it catalyzes the dismutation of superoxide anions. Modeling for the enzymatic mechanism–based crystallographic studies has indicated that during catalysis the copper in the active site is reduced with the substrate $O_2$to yield first O_2 and then H_2O_2. No metal can replace copper because only this metal ion confers catalytic activity to SOD1 . Together, these studies show that copper is a major component of catalytic centers of different redox enzymes and thus its presence is essential for normal physiologic function, such as cellular respiration, free radical defense, synthesis of melanin pigment, connective tissue biosynthesis, and cellular iron metabolism. The importance of the relation between copper and the catalytic feature of redox enzymes is highlighted by studies of SODmutants associated with familial amyotrophic lateral sclerosis(ALS). The discovery of genetic mutations of SOD1 underlying 20–25% of the inherited cases of ALS has generated increased interest in how copper affects SOD1 activity. The initial thought was that neuronal death was associated

with decreased free radical dismutation because of the abnormal enzyme, yet subsequent experiments with SOD1 null mutants in yeast and transgenic mice suggested that the toxic effect of the SOD1 mutation in human ALS cases was in fact related to enhanced activity of some other function and not to lower SOD activity. Hodgson and Fridovich's discovery that SOD1 catalyzes the oxidation of substrates by hydrogen peroxide launched a series of experiments conducted by Wiedau-Pazos *et al* using the formation of electron paramagnetic resonance–detectable hydroxyl adducts of a model substrate [dimethylpyrroline-*N*-oxide (DMPO)] as an index of SOD1 peroxidase activity. The authors discovered that DMPO hydroxyl adduct formation is higher in *SOD1* mutant genes from ALS patients, expressed in yeast. Eliminating copper from either the normal ormutant enzyme abolished the peroxidation of DMPO by hydrogenperoxide. Restoring copper (Cu2+) gradually had a proportional effect in enhancing peroxidase activity. The addition of copper-chelating agents slightly increased peroxidative activityof the normal enzyme while it markedly decreased the activity inthe mutant. From this experiment the authors could not determine whether the chelators remove copper from the active site of the mutant enzyme or bind to the copper, inhibiting the reaction. These studies suggest that copper is fundamental for activity of SOD1 and that altered function of this enzyme is related to the pathologic manifestation in ALS.Copper also acts as an allosteric component of enzymes, probably conferring an appropriate structure for catalytic activity. Protein-bound copper is required for the activity of copper amineoxidases in organisms ranging from yeasts to mammals .These enzymes catalyze the oxidation of various primary biogenicamines and belong to a new class of redox enzymes present in all eukaryotic organisms that contain a peptide-bound TPQ3-(2,4,5-trihydroxyphenyl)-L-alanine (Topa) quinone cofactor. Topaquinone is generated by post translational modification of a specific tyrosyl residue in the highly conserved sequence Asn-Tyr-Asp/Glu. In vitro studies of histamineoxidase indicate that the inactive precursor copper-Topaquinone can be activated by incubation with cupric ions, and that copper reconstituted active enzyme contains the Topaquinone cofactor. This finding[2-4] corroborates that an important aspect of thebiogenesis of these enzymes is a self-catalytic mechanism that involves protein-bound copper and is necessary for complete functional amine oxidase (copper-containing) activity.

COPPER AS AN ESSENTIAL COMPONENT IN GENE EXPRESSION

Metals represent a class of important effector molecules that regulate gene expression in eukaryotic organisms by activation and repression of gene transcription. Studies of copper-regulated transcription in fungi have provided major advancements in the identification of components and mechanisms of action of eukaryotic copper-responsive transcription factors. The role of copper in the function of copper-binding proteins has beens hown for Ace1, Mac1, and Amt1, subsets of regulatory proteins that act physiologically as components of metal-responsive genetic switches. Transcription factors can serve assensors of intracellular copper concentrations or display a regulatory role in a switching process. The mechanism of transcriptional regulation by Ace1 and Amt1 involves copper-induced binding of the factor to a specific upstream activation sequence in the 59- end of the metallothionein promoter. The specific increase of Ace1 DNA-binding activity is achieved through the cooperative formation of a Cu(I)-cysteinylthiolate cluster, which provides the free energy of stabilization for the tertiary fold . Cup9 is another regulatory protein expressed in yeast, this protein acts as a transcriptional factor that regulates the expression of important copper homeostatic genes involved in intracellular partitioning of copper. Metal-responsive elements (MREs) have been found in all eukaryotic metallothionein promoters, which are composed of a series of imperfect repeats containing 13–15 base pairs. However, interesting differences have been found between the yeast and mammalian metallothionein promoter. These differences suggest that the copperdependent transcription regulatory mechanism in mammalian metallothione in species involves a different set of DNA-binding proteins; the protein-protein interaction pattern between these proteins is probably required to form a functional complex of transcription to regulate the expression of metallothionein genes and other target genes. The significance of this difference is unknown but it may be related to different requirements in copperhomeostasis in multicellular organisms[5].A remarkable feature of copper-

dependent transcription factors (Ace1, Amt1, and Mac1) is their association with the expression of other genes related with several physiologic processes. The SOD1 promoter contains a single Ace1 binding site that functions in vivo to coregulate SOD1 and metallothionein transcription in response to copper . In addition, Mac1 has been found to regulate the transcription of two target genes: *FRE1* (encoding a component of plasma membrane associated with both Cu (II) and Fe (III) reduction) and *CTT1* (encoding the cytosolic catalase). Yeast phenotypes of Mac1 loss-of-function mutants can be suppressed by added copper or iron, for respiration-deficient mutants, the phenotype could be rescued only by copper. These studies indicate that the absence of copper in yeast produces drastic effects in cellular processes such as proliferation, growth, and metabolic activity. These effects are related to dysfunction of copper-responsive transcription factors, suggesting a major role of this metal in eukaryotic cellular physiology. There is little information regarding metal-regulated transcription units and their copper-dependent activity in higher eukaryotes. The mouse metallothionein promoter is the most thoroughly studied of the metal-regulated transcription systems in metazoan organisms, and nuclear factors that bind to a metallothione in gene-metal control sequence have been reported. A zinc-finger transcription factor, MTF-I, that binds MREs has been cloned in mice and its activity is postulated to be controlled by a metal-sensitive inhibitor. Interestingly, transient transfection assays of heterogonous chloramphenicol acetyl transferase constructs showed that human MT2A, MT1X, and MT1H promoters can respond to metals, including zinc and copper. More studies of the mechanisms of copper-regulated transcription in mammalian species are necessary, particularly at the cellular and molecular levels, connecting copper-dependent transcription factors and different physiologic cellular processes.

ESSENTIALITY OF COPPER FOR HUMANS

Infants with typical features of copper deficiency (anemia refractory to iron treatment and low plasma concentrations of copper) were first reported in the United Kingdom in 1956. Copper was considered the likely cause for the anemia, but it was not until a series of controlled case studies of copper deficiency in infants recovering from malnutrition was reported by Cordano et al in 1964 that the full spectrum of copper deficiency was shown. Subsequent reports during the 1970s of acquired copper deficiency in low-birth-weight neonates and in infants and children receiving copper-free total parenteral nutrition served to clearly define copper as an essential nutrient for human infants. Concomitantly, tissue copper deficiency as a result of a rare inherited defect of copper transport, Menkes syndrome, was also recognized. Copper deficiency is more commonly an acquired condition induced by the imbalance between need and dietary copper supply. We will first address the acquired conditions leading to copper deficiency and then briefly refer to the inherited metabolic defect.

FACTORS CONDITIONING COPPER DEFICIENCY IN HUMANS

Copper deficiency is more frequent in preterm infants, especially those with very-low birth weights, because of their reduced copper stores at birth given the smaller relative size of the liver and higher requirements determined by their high growth rate compared with full-term infants . Infants fed cow milk–based diets exclusively are more prone to develop copper deficiency than infants who are breast-fed because of the low copper content of cow milk and limited absorption of this mineral in cow milk. In contrast, breast-fed infants absorb more copper, which may be due to the lower casein content of human milk or to factors associated with human milk that enhance copper absorption**.** Copper deficiency has been reported in subjects with mal absorption syndromes—such as celiac disease, tropical and nontropicalsprue, cystic fibrosis, and short-bowel syndrome— resulting from intestinal resection. Increased gastrointestinal copper losses usually explain the occurrence of copper deficiency in malabsorption syndromes. Copper deficiency should be suspected in infants with prolonged or recurrent diarrheal episodes, abnormal bile loss, intestinal resections, or loss of intestinal contents from intestinal fistula. High oral intakes of zinc and iron decrease copper absorption and may predispose to copper deficiency. This phenomenon is used as a therapeutic strategy in Wilson disease, in which a high zinc intake (40–50 mg/d) has been shown to lower copper absorption. Copper deficiency also has been documented in subjects receiving penicillamine, or other

cation-chelating agents, or high doses of oral alkalis, which enhance copper losses.

During total parenteral nutrition, insufficient copper supply will occur unless subjects are supplemented with trace elements, including copper. In subjects with impaired biliary secretion orwith cholestasis, copper supplementation should be reduced tocompensate for the limited losses. In patients with impaired liver function, copper supplementation should be withheld unless biochemical signs of copper deficit are documented. Enteral alimentation products also need to contain adequate copper to meet the needs of individual patients. On the basis of published information, the most common cause of copper deficiency is an insufficient copper supply during the nutritional recovery of malnourished children. Several factors are frequently associated with copper deficiency in these infants: low birth weight, short duration of breast-feeding, cow milk consumption, consumption of highly refined carbohydrate based diets, increased losses of nutrients as a result of diarrheal disease, and frequent infections. During nutritional recovery, these infants grow at 5–10 times the normal rate for their age, thus increasing the requirements imposed by growth. This is yet another risk factor contributing to the 30–40% prevalence of biochemical signs of copper deficiency reported in these children. Castillo-Durán et al evaluated the magnitude of copper losses in 14 infants during an acute diarrheal episode requiring hospitalization. The results were compared with those obtained in 15 matched control infants. Fecal losses in the diarrhea group were two times greater than those in the control group. The diarrhea group was in negative copper balance up to 7 d after hospital admission. Copper losses were directly related to fecal weight. Furthermore, Rodriguez et al compared the copper status of 19 children with chronic diarrhea with that of two control groups (19 healthy and 11 malnourished children). Plasma copper concentrations were 30% lower and hair copper content decreased three to fourfold in the chronic diarrhea group relative to the control groups. Low copper intakes during recovery from diarrhea may further limit copper nutriture[6].

REGULATION OF COPPER NUTRITURE IN EARLY LIFE

The fetus is fully dependent on the maternal copper supply and during pregnancy, copper and ceruloplasmin serum concentrations rise significantly. Pregnancy is associated with increased copper retention, which may be due in part to decreased biliary copper excretion induced by hormonal changes typical during pregnancy. Serum copper after the first trimester of pregnancy rises from a mean of <15.7 mmol/L (<100 mg/dL) to double this value during the last trimester. Maternal concentrations during the latter half of gestation are five to seven times concentrations measured in the cord. The fetus accumulates copper at a mean rate of 50 mg ?kg21 ?d21 principally over the latter half of pregnancy; > 50% of the copper is stored in the liver, mainly in metallothionein . Fetal serum copper and ceruloplasmin are low and remain stable for most of the third trimester, despite active accretion in fetal tissues. The increase in fetal liver stores is due both to increased liver size and to a higher copper concentration per unit of liver weight .The brain is the second site for copper accumulation during fetal life; in Menkes syndrome, the brain copper content is extremely low and the basal ganglia and cerebellum are particularly affected. Copper uptake into the fetal compartment is dependent on placental carrier-mediated copper transport from ceruloplasmin. Other copper-binding complexes such as albumin or histidine bound copper can also contribute to the carrier-mediated system (81). By the end of gestation, a full-term fetus will have accumulated close to 15 mg Cu, 9 mg of which will be in the liver. After birth, the concentration of copper in the liver drops steadily because the copper supply from the diet rarely meets the requirements for rapid growth typical of this stage. In addition, biliary secretion increases steadily during early life, augmenting fecal copper losses. Plasma copper and ceruloplasmin rise after birth, reaching adult values by 6 mg of age. Mean serum copper on day 7 ranges from 4.7 to 7.9 mmol/L (30 to 50 mg/dL), depending on gestational age, and ceruloplasmin concentrations range from 3.1 to 4.7 mmol/L (20 to 30 mg/dL) accordingly. Traditional mineral balance studies conducted in preterm or full term infants fed cow milk–based diets or unfortified pasteurized human milk indicated that most infants were in negative copper balance or at best in marginally positive balance .These observations, coupled with the lack of an effect of copper supplementation in determining a faster rise in serum copper and ceruloplasmin after birth, suggested that infants do not absorb copper well. Changes in intake during the first months of life up to four times the estimated requirements do not affect copper or ceruloplasmin concentrations. More recent studies in infants

fed modern artificial formula or unpasteurized human milk, which used a combination of chemical-balance and stable-isotope-tracer (65Cu) methods, indicated that preterm infants can absorb copper; moreover, if sufficient copper is provided they can retain enough copper to meet the requirements imposed by growth. Infants fed preterm human milk absorb close to 60% of their intake whereas those receiving premature formula will absorb only 15% ; the respective values for true 65Cu absorption are 67% and 39%, respectively. Fecal copper losses are higher with increasing intake whereas percentage absorption decreases. The interpretation of these results is confounded by the fact that human milk provides much less copper and is associated with significantly higher copper absorption. The absolute retention of copper in infants fed human milk approached the expected retention based on in utero accretion data. The excretion of copper in the feces was directly correlated with fat and nitrogen fecal losses; thus, if fat is mal absorbed, copper is lost .This finding may explain some of the differences between resultsof modern studies of copper balance and those of older studies using pasteurized human milk or old artificial formulas, both of which are associated with higher fecal fat losses. In summary, modern copper balance studies show that copper balance in early life is affected by the type of feeding and the amount of copper supplied. If insufficient copper is supplied or if the copper supplied is poorly absorbed, deficiency may occur. Infants adjust copper absorption, reducing absorption at higher intakes and increasing absorption at lower intakes. The elevated hepatic copper stores present at birth and the low ceruloplasmin concentrations early in life suggest that infants cannot adjust to high copper intakes. Some have suggested that even minimal copper loads may be associated with liver toxicity, on the basis of the rare occurrence of Indian childhood cirrhosis and idiopathic copper toxicosis; for idiopathic copper toxicosis,the incidence is estimated to range from 1 in 100 000 to 1 in 1 000 000. According to these findings, copper should be eliminated from the diet early in life. This is in sharp contrast with the common occurrence of biochemical signs of copper deficiency when copper is not added to infants' diets. Low birth weight and protein energy malnutrition affect a sizable proportion of children throughout the world. It has been estimated that > 10% of children are born with low birth weight .Thus, the risk of deficiency must be weighed against the risk of potential toxicity, which affects an extremely few selected individuals who most likely carry a genetic predisposition for copper accumulation. Recommendations for infant feeding and total copper exposure, including copper in potable water, should take into account that copper are an essential nutrient and that toxicity is possible if the copper load exceeds tolerance. A range of safe intakes should be defined for the general population, including a lower safe intake and an upper safe intake, to ensure that for most of the population both deficiency and toxicity are prevented. When setting guideline values for populations, the special needs of genetically susceptible individuals should be considered only if there is public health significance.

CONCLUSION

The imine ligands form very stable complexes with copper, and can efficiently completely for this metal ion towards the specific N-terminal binding sites of the protein ,consisting of the amino acid sequence Asp-Ala-His. Copper and its compounds are widely used as medicine for long years including the treatment of chest wound and purifying drinking water .Research has indicated that Cu and its compound helps prevent inflammation in arthritis and similar diseases .Research is going on into anti-ulcer and anti-inflammatory medicine containing copper and its compound and used in radiology and for treating convulsion and epilepsy.

REFERENCES

1. **Hart, E.B., Steenbock, H.,Waddell, J. and Elvehjem, C.A.1928**. Iron nutrition. VII. Copper is a supplement to iron for hemoglobin building in the rat. J.BiolChem; 77:797–812.
2. **Mills, E.S. 1930**. The treatment of idiopathic (hypochromic) anemia with iron and copper. Can Med Assoc J; 22:175–8.
3. **Josephs, H.W. 1990**. Treatment of anemia of infancy with iron and copper.Bull Johns Hopkins Hosp 1931;49:246–58.
4. **Gacheru, N., Trackman, P.C., Shah, M.A. 1990**. Structural and catalytic properties of copper in lysyl oxidase. J BiolChem;265:19022–7.
5. **Krebs, C.J., Krawetz, S.A. 1993.** Lysyl oxidase copper-talon complex: a model. BiochimBiophysActa;1202:7–12.
6. **Palumbo, M., d'Ischia, G., Misuraca, G., Carratu, L., Prota, G. 1990.** Activationby ferrous ions. BiochimBiophysActaof mammalian tyrosinase ;1033:256–60.

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